

AI Intern

Week 1 – Foundations of Artificial Intelligence



BASIC NEURAL NETWORK FROM SCRATCH

Description: Build a simple neural network to classify digits using NumPy only (no ML libraries).

Challenges:

- Implement forward propagation & activation functions
- Calculate loss and accuracy
- Visualize results

Tech Stack: Python, NumPy, Matplotlib

RULE-BASED CHATBOT

Description: Create a conversational chatbot using pattern-matching rules.

Challenges:

- Intent recognition using keyword mapping
- Add fallback responses
- Support 3–4 conversation topics

Tech Stack: Python, Regex/NLP basics

AI-BASED TIC-TAC-TOE GAME

Description: Create a game opponent using Minimax algorithm.

Challenges:

- Implement optimal decision-making
- UI using CLI or simple GUI
- Track wins/losses

Tech Stack: Python, Minimax Algorithm

***Complete any 1 Task**